

Illumina

NASDAQ : ILMN \$21.9B mkt cap · 151M sh · \$1.16B cash, \$1.99B net debt · ~\$0.83B net debt; ~151M shares; ~\$700M/yr buyback (no dividend); Revenue
· Mature- \$4.34B FY25 (~flat 4yrs; Q1'26 +4.8% re-accel); ~23% non-GAAP op margin, ~70% gross, ~\$0.9-1.0B FCF; Rallied ~80% off lows on the
Company DCF clinical/NovaSeq-X turnaround; ~5x EV/sales, ~22x FCF

\$144.83

THE CENTRAL QUESTION

Is Illumina's ~80% rally + ~5x EV/sales pricing a durable clinical/NovaSeq-X re-acceleration — or is flat revenue + low-cost-entrant (Ultima/Element) + Roche erosion the reality?

Illumina trades at ~\$22B (~\$145/sh, up ~80% off the lows) — ~5x EV/sales, ~22x FCF on \$4.34B revenue (flat for four years, Q1'26 +4.8%), ~70% gross margin, ~\$1B FCF. The mature DCF asks whether the NovaSeq-X consumables pull-through + clinical mix (>65% of sequencing consumables, +20% ex-China) re-accelerate the franchise toward a ~26% margin, against low-cost entrants (Ultima/Element \$100 genome), Roche's 2026 SBX re-entry, China decline, and NIH cuts. The finding: the rally has priced the re-acceleration — only the bull-or-better clears spot; the base (modest re-accel) sits ~27% below. The TXG mirror.

PROBABILITY-WEIGHTED EXPECTED VALUE

\$101.53 expected fair value today
-29.9% vs spot \$144.83

FORWARD COMPOUNDED VALUE

+5y	+10y	+15y	+20y
\$156.27	\$240.77	\$371.34	\$573.33
1.08x spot	1.66x spot	2.56x spot	3.96x spot

WEIGHTING RATIONALE

Ultra Bear 12% Entrants + China + NIH crush; ~\$40 (-73%).

Bear 30% Flat; clinical offsets; ~\$92 (-37%).

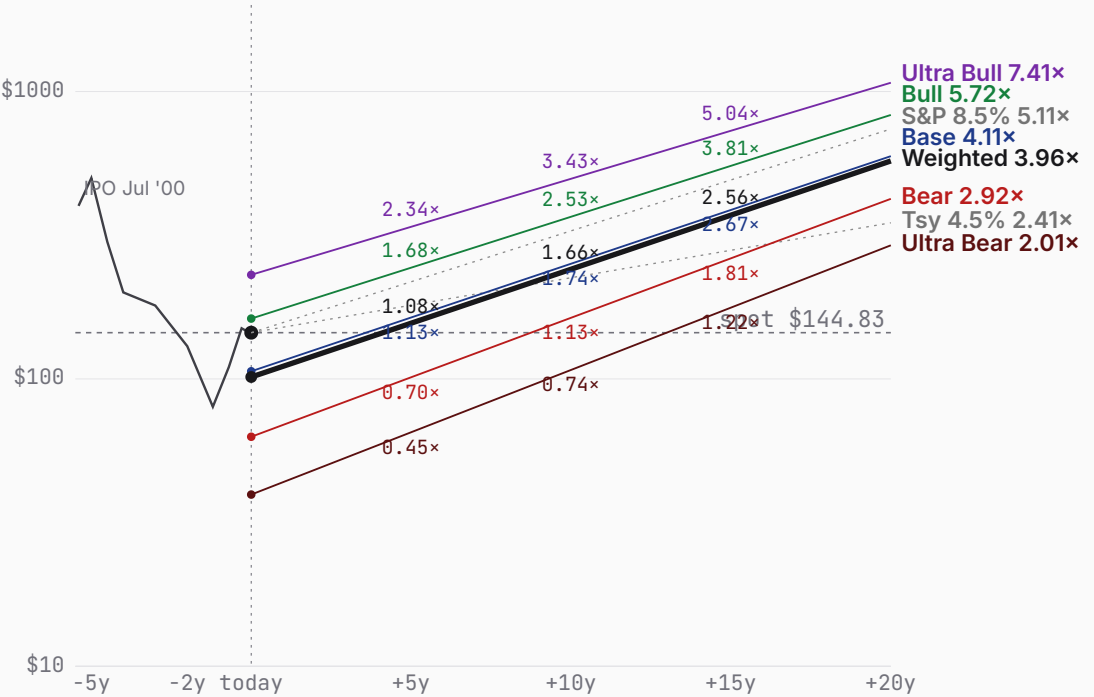
Base 35% Modest re-accel; ~26% margin; ~\$106 (-27%).

Bull 18% Clinical compounds; multiomics; ~\$162 (+12%).

Ultra Bull 5% Sequencing super-cycle; ~\$230 (+59%).

Bull (18%) + Ultra Bull (5%) together contribute \$40.70 — 40% of the \$101.53 expected value despite 23% combined probability. Today's spot (\$144.83) sits 43% above the weighted expected; forward-weighted value crosses spot between +5y and +10y.

PRICE + PROJECTIONS · HISTORICAL THROUGH PROBABILITY-WEIGHTED FORWARD



ULTRA BEAR	\$39.59	BEAR	\$62.91	BASE	\$106.30	BULL	\$162.15	ULTRA BULL	\$230.21
	-72.7% vs spot		-56.6% vs spot		-26.6% vs spot		+12.0% vs spot		+59.0% vs spot
CAGR -1.0% WACC 10.5% CAGR -1.0% Prob 12%		CAGR +1.4% WACC 10.0% CAGR +1.4% Prob 30%		CAGR +4.6% WACC 9.0% CAGR +4.6% Prob 35%		CAGR +7.2% WACC 8.5% CAGR +7.2% Prob 18%		CAGR +10.0% WACC 8.0% CAGR +10.0% Prob 5%	
Low-cost entrants + China + NIH crush.		Flat; clinical offsets erosion; rally unwinds.		Modest re-accel; NovaSeq-X pull-through; ~26% margin.		Clinical compounds; multiomics; margin >26%.		Sequencing super-cycle; dominant + multiomics.	

PROBABILITY WEIGHTS

Ultra Bear 12% Bear 30% Base 35% Bull 18% Ultra Bull 5% · Weighted expected \$101.53 (-29.9% vs spot)

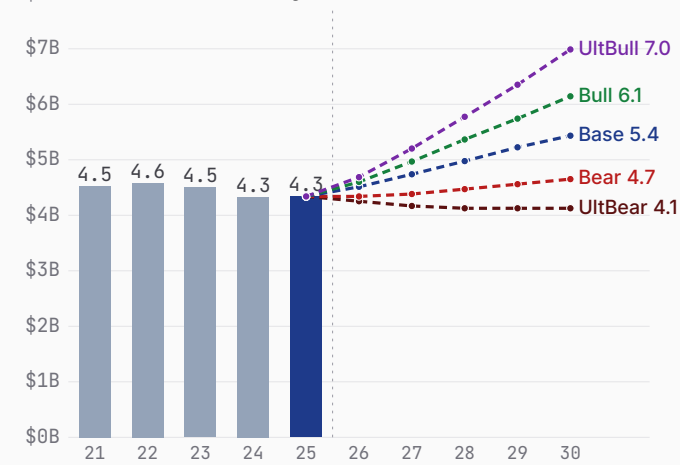
ULTRA BEAR	\$39.59	BEAR	\$62.91	BASE	\$106.30	BULL	\$162.15	ULTRA BULL	\$230.21
	-72.7% vs spot		-56.6% vs spot		-26.6% vs spot		+12.0% vs spot		+59.0% vs spot
Probability 12%		Probability 30%		Probability 35%		Probability 18%		Probability 5%	
Low-cost entrants + China + NIH crush.		Flat; clinical offsets erosion; rally unwinds.		Modest re-accel; NovaSeq-X pull-through; ~26% margin.		Clinical compounds; multiomics; margin >26%.		Sequencing super-cycle; dominant + multiomics.	
WHY 12%		WHY 30%		WHY 35%		WHY 18%		WHY 5%	
The bear: the SBS patent estate ages, low-cost entrants design around it, and the academic base shrinks. 12% weight on real erosion.		Flat revenue (the FY22-25 reality) with clinical offsetting erosion. 30% weight — the rally priced more than this.		NovaSeq-X pull-through + clinical + the ~26% margin target, consistent with Q1'26's +4.8% and raised guide. 35% as the central outcome — but priced in.		The clinical-led re-acceleration is durable + multiomics adds growth. ~18%; the bull the rally is discounting in.		Sequencing super-cycle + ILMN defends its Power. ~5%, the asymmetric upside.	
WHAT HAPPENS		WHAT HAPPENS		WHAT HAPPENS		WHAT HAPPENS		WHAT HAPPENS	
Ultima/Element (\$100 genome) compress consumables pricing/share, Roche's SBX re-entry bites, China keeps declining, and NIH cuts freeze research demand. Revenue declines and the consumables annuity erodes.		Clinical growth roughly offsets China + entrant erosion, leaving revenue ~flat at a recovering margin; the franchise is dominant but ex-growth.		The modal path: the NovaSeq-X installed base matures into a consumables acceleration, clinical compounds, and the margin marches to ~26% (management's 2027 target). Revenue grows ~4-5%.		Clinical sequencing compounds high-single-digits, multiomics/proteomics (SomaLogic) cross-sells, and the margin pushes past 26% as NovaSeq-X pull-through accelerates in 2H'26+.		A genomics super-cycle (clinical at scale + multiomics) re-accelerates ILMN to low-double-digits at ~30% margins, defending its ~80% share against entrants.	
DCF ~\$40 (-73%) — a shrinking franchise losing its pricing Power earns a distressed multiple.		DCF ~\$92 (-37%) — well below the rallied price; flat revenue doesn't justify ~5x sales.		DCF ~\$106 (-27%) — even a successful modest re-acceleration doesn't justify the ~80% rally; the entry has run ahead of the fundamentals.		DCF ~\$162 (+12%) — the durable re-acceleration the rally is pricing actually materializes.		DCF ~\$230 (+59%) — the dominant-platform-re-rates outcome; ~5% probability.	

Illumina business snapshot

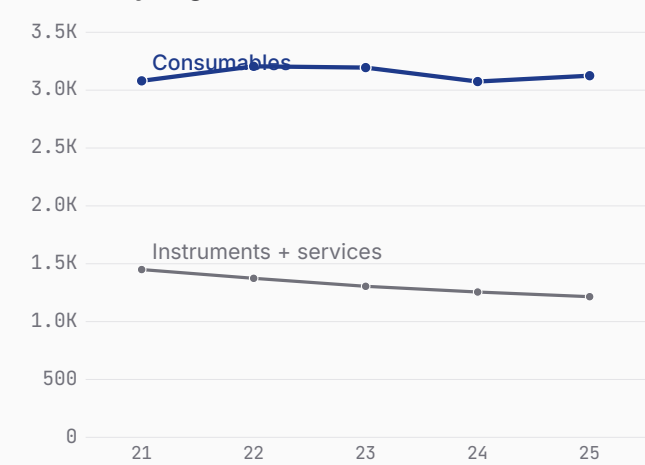
Bear \$62.91 Base \$106.30 Bull \$162.15

FY21-FY25 history + FY26-FY30 scenario projections · calendar fiscal year · FY2025 10-K + Q1'26 (Core Illumina, post-GRAIL)

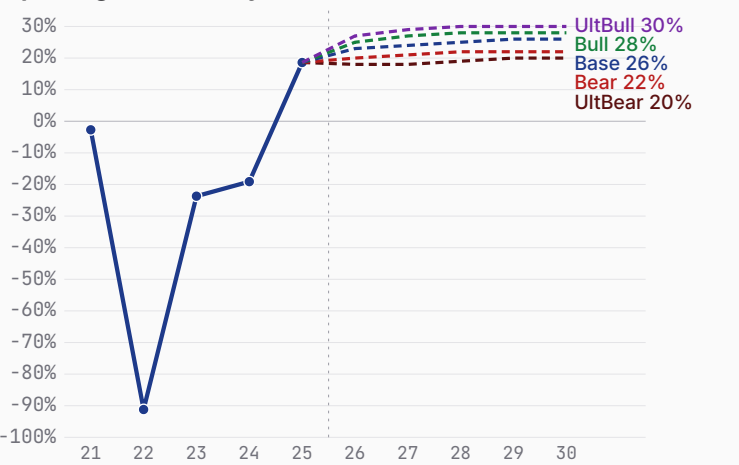
Revenue (\$B) — history + scenarios to FY30



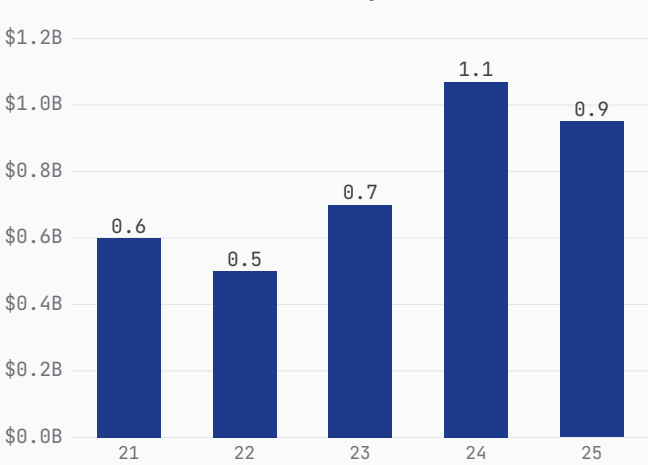
Revenue by segment (\$M)



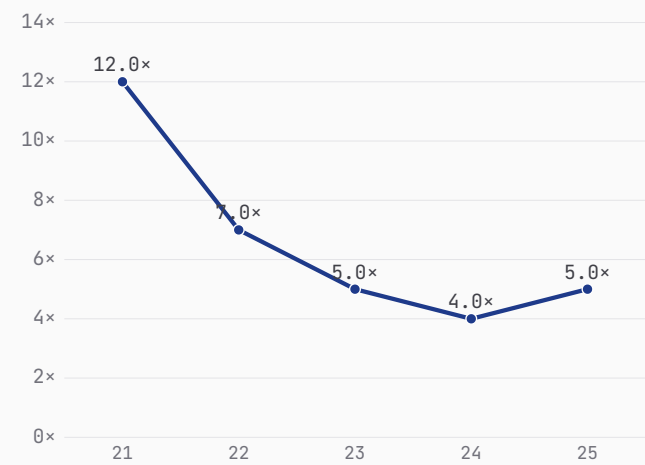
Op margin — history + scenarios



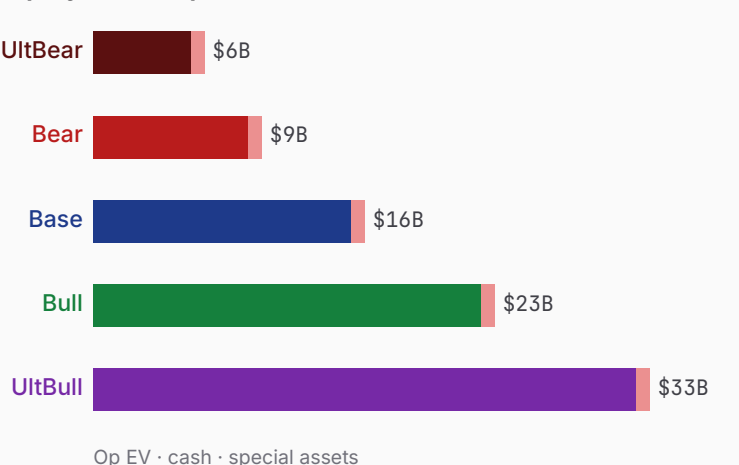
Free cash flow (\$B) — history



EV / Revenue — historical



Equity build (Op EV - net debt)



Sources: Illumina FY2025 results (Core revenue \$4.34B, ~23% non-GAAP op margin, ~\$1B FCF, net debt ~\$0.83B), Q1'26 (+4.8%, raised guide). GRAIL divested Jun 2024 → Core-only. Revenue as a growth-rate path off FY25; FCF = revenue x FCF margin; Gordon terminal. EV/sales vs life-science-tools peers.

Illumina 7 Powers · the moat, scored

Bear \$62.91 Base \$106.30 Bull \$162.15

Arena. DNA sequencing — Illumina (~80% share, NovaSeq-X installed base, ~70%-margin consumables annuity) vs Ultima, Element, Roche (SBX, 2026), PacBio, Thermo, MGI/Complete Genomics; defending a scale + switching Power against low-cost entrants. (Supplies 10x Genomics/TXG; the two are IP litigants.)

POWER AUDIT
dominant-power durability: medium

POWER AUDIT · 7 POWERS

Scale Economies · thesis leans here	~80% share + huge installed base → manufacturing/R&D cost advantage on consumables; the dominant Power.
Network Economies	Published-protocol/citation network is mild.
Counter-Positioning	Incumbent; low-cost entrants counter-position on price-per-genome.
Switching Costs	Validated clinical assays + DRAGEN pipelines on Illumina chemistry won't re-validate cheaply (regulatory friction).
Branding	The 'gold standard' in clinical NGS.
Cornered Resource	Sequencing-by-synthesis patent estate — established but contested/aging (litigated vs Element/BGI).
Process Power	Chemistry/manufacturing learning curve.

PEER SET

Ultima Genomics	· private
UG 100, ~\$80-100 genome (high-throughput) — the low-cost price attack.	
Element Biosciences	· private
AVITI + Vitari (\$100 30x genome); mutual IP litigation with Illumina.	
Roche (SBX)	· diversified
Re-entering with sequencing-by-expansion (2026); diagnostics channel — the credible long-term threat.	
PacBio / Thermo / MGI	· varies
Long-read niche / Ion Torrent / China (MGI blocked from US by BIOSECURE).	

THREAT VECTORS · FALSIFIERS

- Scale Economies · Ultima / Element (\$100 genome)
falsifier: Low-cost entrants take material high-throughput share; consumables pricing compresses.
- Cornered Resource · Roche SBX (2026)
falsifier: Roche's SBX gains clinical traction, breaking the SBS franchise.
- Switching Costs · NIH / China
falsifier: NIH cuts + China decline shrink the academic installed base the annuity depends on.

COMPETITIVE READ

Illumina's scale + switching-cost Power is genuine and durable — ~80% share, >90% of clinical genomics, a ~70%-margin consumables annuity, and clinical labs that won't cheaply re-validate. But the Power is being audited: revenue was flat for four years, China collapsed, NIH froze research demand, low-cost entrants (Ultima/Element \$100 genome) attack pricing, and Roche's SBX re-enters in 2026. The ~80% rally to ~5x EV/sales has priced the NovaSeq-X/clinical re-acceleration; only the bull-or-better clears spot, and the modal base sits ~27% below — the entry, not the franchise, is the problem (the TXG mirror).

Illumina
show your work

Bear \$62.91 Base \$106.30 Bull \$162.15 · Weighted \$101.53 (-29.9%)

ULTRA BEAR					BEAR					BASE					BULL					ULTRA BULL				
\$39.59					\$62.91					\$106.30					\$162.15					\$230.21				
ASSUMPTIONS					ASSUMPTIONS					ASSUMPTIONS					ASSUMPTIONS					ASSUMPTIONS				
5y revenue CAGR (%)					-1.0					5y revenue CAGR (%)					+1.4					5y revenue CAGR (%)				
Year-5 op margin (%)					20.0					Year-5 op margin (%)					22.0					Year-5 op margin (%)				
WACC (%)					10.5					WACC (%)					10.0					WACC (%)				
Terminal growth (%)					1.0					Terminal growth (%)					2.0					Terminal growth (%)				
5y revenue CAGR (%)					-1.0					5y revenue CAGR (%)					+1.4					5y revenue CAGR (%)				
Starting revenue (\$B)					4.34					Starting revenue (\$B)					4.34					Starting revenue (\$B)				
Scenario probability (%)					12					Scenario probability (%)					30					Scenario probability (%)				
5-YEAR DCF · \$B					5-YEAR DCF · \$B					5-YEAR DCF · \$B					5-YEAR DCF · \$B					5-YEAR DCF · \$B				
FY	Rev	Margin	FCF	PV FCF	FY	Rev	Margin	FCF	PV FCF	FY	Rev	Margin	FCF	PV FCF	FY	Rev	Margin	FCF	PV FCF	FY	Rev	Margin	FCF	PV FCF
FY26	4.25	+18%	+0.68	+0.61	FY26	4.34	+20%	+0.78	+0.71	FY26	4.51	+23%	+0.90	+0.83	FY26	4.60	+25%	+1.01	+0.93	FY26	4.69	+27%	+1.13	+1.04
FY27	4.17	+18%	+0.63	+0.51	FY27	4.38	+21%	+0.79	+0.65	FY27	4.74	+24%	+0.99	+0.84	FY27	4.97	+27%	+1.19	+1.01	FY27	5.20	+29%	+1.35	+1.16
FY28	4.13	+19%	+0.62	+0.46	FY28	4.47	+22%	+0.85	+0.64	FY28	4.98	+25%	+1.09	+0.85	FY28	5.37	+28%	+1.34	+1.05	FY28	5.78	+30%	+1.56	+1.24
FY29	4.13	+20%	+0.66	+0.44	FY29	4.56	+22%	+0.87	+0.59	FY29	5.23	+26%	+1.15	+0.81	FY29	5.74	+28%	+1.44	+1.03	FY29	6.35	+30%	+1.78	+1.31
FY30	4.13	+20%	+0.66	+0.40	FY30	4.65	+22%	+0.88	+0.55	FY30	5.43	+26%	+1.20	+0.78	FY30	6.14	+28%	+1.54	+1.02	FY30	6.99	+30%	+1.96	+1.33
Σ PV explicit FCF					+2.43					Σ PV explicit FCF					+4.10					Σ PV explicit FCF				
+ PV terminal (g 1.0%)					4.26					+ PV terminal (g 2.0%)					12.25					+ PV terminal (g 3.0%)				
EQUITY BUILD · \$B & PER SHARE					EQUITY BUILD · \$B & PER SHARE					EQUITY BUILD · \$B & PER SHARE					EQUITY BUILD · \$B & PER SHARE					EQUITY BUILD · \$B & PER SHARE				
Operating EV					\$6.69B					Operating EV					\$10.14B					Operating EV				
+ Cash & investments					\$1.16B					+ Cash & investments					\$1.16B					+ Cash & investments				
– Net debt					\$1.99B					– Net debt					\$1.99B					– Net debt				
= Equity value					\$5.86B					= Equity value					\$9.31B					= Equity value				
FY30 shares (M)					148					FY30 shares (M)					148					FY30 shares (M)				
= Expected per share					\$39.59					= Expected per share					\$62.91					= Expected per share				
FUTURE FAIR VALUE · IF SCENARIO PLAYS OUT					FUTURE FAIR VALUE · IF SCENARIO PLAYS OUT					FUTURE FAIR VALUE · IF SCENARIO PLAYS OUT					FUTURE FAIR VALUE · IF SCENARIO PLAYS OUT					FUTURE FAIR VALUE · IF SCENARIO PLAYS OUT				
+5y	+10y	+15y	+20y		+5y	+10y	+15y	+20y		+5y	+10y	+15y	+20y		+5y	+10y	+15y	+20y		+5y	+10y	+15y	+20y	
\$65.22	\$107.45	\$177.02	\$291.63		\$101.32	\$163.17	\$262.79	\$423.23		\$163.56	\$251.65	\$387.20	\$595.75		\$243.82	\$366.62	\$551.27	\$828.92		\$338.25	\$497.01	\$730.27	\$1,073.00	
0.45×	0.74×	1.22×	2.01×		0.70×	1.13×	1.81×	2.92×		1.13×	1.74×	2.67×	4.11×		1.68×	2.53×	3.81×	5.72×		2.34×	3.43×	5.04×	7.41×	

Illumina

People · Opportunity · Context · Deal

Bear \$62.91 Base \$106.30 Bull \$162.15

Underwrite the position the way a venture investor underwrites a round — across all four legs, public or private. **People** is the leg scored here (observable only); Opportunity, Context and Deal cross-reference the rest of the memo.

PEOPLE · WHO RUNS / OWNS / FUNDS IT

Jacob Thaysen (CEO since Sept 2023)

2%insider ownership

LOWkey-person risk

Capital allocation (realized)
Dominated by the GRAIL episode — a major observable value-destruction: bought GRAIL (~\$8B, 2021) and CLOSED it without regulatory clearance, drawing an EU prohibition + an unprecedented €432M "gun-jumping" fine (later voided on jurisdiction, 2024) and a forced spin-off (June 2024 → GRAL) that took a ~\$4B impairment + a ~\$1B recapitalization. No dividend; small buybacks (\$1.5B authorized Apr 2026); R&D-heavy.

Incentive alignment
Comp redesigned after a pay revolt: CEO target equity \$10M is 70% PSUs / 30% RSUs, PSUs vesting on 3-year relative TSR (the make-whole grant made explicitly 70% TSR-contingent in response to stockholder feedback). Base \$1M, target bonus 125%.

GOVERNANCE FLAGS

- single-class, declassified board (annual elections)
- 2023 Icahn proxy fight: chairman John Thompson ousted; Icahn won one board seat
- say-on-pay FAILED 2023 → recovered ~86.5% (2024) after the comp redesign
- board/chair overhaul since (Gottlieb non-exec chair 2025; Corvex Meister joined; Icahn fully exited)

PEOPLE READ

The cohort's governance-turnaround story — and the weakest recent record: a board that closed an \$8B deal without clearance (forced unwind, ~\$4B impairment, an EU fine), failed say-on-pay (2023), and lost a proxy seat to Icahn. Being repaired — new CEO, independent chair, redesigned TSR-linked pay, say-on-pay recovered, activist oversight — but the track record is the observable flag.

THE FOUR LEGS

People

observable composite · 1–5

Opportunity

The scenario distribution · the Page-3 business snapshot · the Arthur Indicator (ILMN 7.08)

Context

The 7 Powers analysis (Power Audit)

Deal

Open-market purchase = the deal: entry price vs. the scenario distribution (the ~30% finding) + position sizing.

Illumina supporting analysis · disclaimers · glossary

Bear \$62.91 Base \$106.30 Bull \$162.15

PUSHBACK · WHY THE BASE CASE IS TOO HARSH

- 1

~80% share, >90% of clinical genomics.
Dominant installed base + consumables lock-in (validated clinical assays don't re-validate cheaply) — a real scale + switching Power.
- 2

~70% gross margin, ~\$1B FCF.
A razor/blade annuity (consumables ~72% of revenue) with a clear path to a ~26% operating margin (2027 target).
- 3

NovaSeq-X pull-through is coming.
~890 installed (end-FY25); ~6-month pull-through lag → a 2H'26 consumables acceleration the base is just starting to price.
- 4

Clinical mix is the durable engine.
>65% of sequencing consumables, +20% ex-China — less funding-exposed than the academic/research side.
- 5

GRAIL overhang is gone.
Divested Jun 2024; Core-only reporting, EU fine/impairments behind it; new CEO running the cost-out + clinical pivot.

FALSIFICATION TRIGGERS

Bull validation

total revenue growth sustains >6% · NovaSeq-X consumables accelerate (2H'26) · non-GAAP op margin reaches ~26% · clinical stays +20% ex-China

Bear validation

revenue stays flat-to-down · Ultima/Element take material share · Roche SBX gains clinical traction · China/NIH drag deepens

Reframe needed

if a low-cost entrant or Roche SBX structurally breaks the SBS consumables-pricing Power, re-rate toward a commoditized-sequencing multiple

DISCLAIMERS · PLEASE READ BEFORE USING THIS DOCUMENT

Not investment advice.

This document is produced for entertainment and educational purposes only by an individual amateur investor. "Alameda Research 2: Electric Boogaloo" is the unincorporated personal concept of one person and is not a registered investment management entity. Nothing herein constitutes investment advice, a recommendation, or a solicitation to buy or sell any security.

Not a professional.

The author is not a registered investment advisor, broker-dealer, CFA charterholder, certified financial planner, or licensed financial professional of any kind. The author has no formal training in equity research or capital markets. No fiduciary relationship is created by reading this document.

AI-assisted analysis.

Substantial portions of this analysis were produced with the assistance of large language models. LLMs are known to fabricate facts, miscalculate numbers, hallucinate sources, and present incorrect information with high apparent confidence. Every figure, claim, and projection in this document should be independently verified before acting on it.

Forward-looking statements.

Scenarios, valuations, projections, and any forward-looking statements involve substantial assumptions and uncertainty. Actual results may differ materially from those projected. Past performance is not indicative of future results. The model is a model; reality is not.

Author may hold positions.

The author may own, intend to acquire, or hold short exposure to \$ILMN or related securities at any time. Positions may change without notice and without an update to this document. Do not assume the author's positions match the tone of the analysis.

Do your own research.

Do not make investment decisions on the basis of this document. Consult a licensed financial professional and read the company's official SEC filings (10-K, 10-Q, 8-K, proxy) before forming any investment view. If you wouldn't trust your retirement to a chatbot, don't trust this either.

GLOSSARY · CONCEPTS REFERENCED IN THE NARRATIVE

Consumables (razor/blade) — Reagents/flow-cells that run on the installed base (~72% of revenue, ~70% margin) — the recurring annuity and the moat.

NovaSeq X — Illumina's flagship high-throughput sequencer; its installed-base ramp drives the consumables pull-through (the re-acceleration mechanism).

Clinical mix — >65% of sequencing consumables, +20% ex-China — the durable, less-funding-exposed demand engine vs academic/research.

Low-cost entrants — Ultima (~\$80-100 genome) + Element (Vitari) + Roche SBX (2026) attacking the sequencing-by-synthesis franchise on price.

FCF margin — Free cash flow / revenue (~21% FY25); the mature-DCF lever as the margin marches to the ~26% 2027 target.